

Rule	Conf.	RoBERTa-wBCE			RoBERTa		
		Acc.	CA@k		Acc.	CA@k	
			.10	.01		.10	.01
birthYear(a,c) $\wedge$ deathYear(b,d) $\wedge$ $>(c,d) \rightarrow \text{negspouse}(a,b)$	.990	.995	.993	.993	.970	.490	.486
birthYear(b,d) $\wedge$ foundYear(a,c) $\wedge$ $<(c,d) \rightarrow \text{negfounder}(a,b)$	.990	.928	.927	.927	.908	.486	.456
spouse(c,a) $\wedge$ parent(b,c) $\rightarrow \text{negspouse}(a,b)$	.923	.974	.963	.747	.875	.491	.279
relative(a,c) $\wedge$ spouse(b,c) $\wedge$ child(b,a) $\rightarrow \text{relative}(a,b)$	.860	.922	.844	.801	.866	.342	.146
parent(c,a) $\wedge$ child(b,c) $\rightarrow \text{spouse}(a,b)$	.825	.944	.828	.444	.842	.342	.146
publisher(c,b) $\wedge$ subsequentWork(c,a) $\rightarrow \text{publisher}(a,b)$	.721	.909	.834	.765	.905	.358	.219
successor(b,a) $\rightarrow \text{negspouse}(a,b)$	.718	.972	.896	.693	.949	.369	.313
child(c,b) $\wedge$ relative(c,a) $\rightarrow \text{negchild}(a,b)$	.644	.935	.880	.693	.905	.310	.303
child(c,b) $\wedge$ spouse(a,c) $\rightarrow \text{negrelative}(a,b)$	.562	.920	.907	.608	.915	.255	.250
relation(a,b) $\rightarrow \text{negchild}(a,b)$	.549	.904	.886	.737	.902	.371	.366
child(c,b) $\wedge$ spouse(c,a) $\rightarrow \text{child}(a,b)$	.492	.901	.827	.422	.658	.223	.107
knownFor(b,a) $\rightarrow \text{founder}(a,b)$	.387	.882	.601	.477	.839	.372	.215
founder(c,b) $\wedge$ publisher(c,a) $\rightarrow \text{negfounder}(a,b)$	.246	.886	.795	.665	.802	.311	.297
publisher(a,c) $\wedge$ parentCompany(b,c) $\rightarrow \text{negpublisher}(a,b)$	.235	.812	.748	.643	.811	.313	.271
successor(c,a) $\wedge$ spouse(c,d) $\wedge$ successor(d,b) $\rightarrow \text{spouse}(a,b)$	.221	.927	.738	.628	.761	.248	.215
relative(a,c) $\wedge$ parent(c,b) $\rightarrow \text{child}(a,b)$	.135	.841	.704	.552	.727	.227	.182

Table 2: Evaluation results for single-rule models.

Test	Size	F1	CA@.15	CA@.1	CA@.05
r_1	1.6k	.990	.987	.986	.954
r_2	1.6k	.999	.997	.996	.946
r_3	1.6k	.995	.994	.994	.992
r_4	1.6k	.990	.989	.988	.935
r_5	1.6k	1	.999	.998	.979
U=2	20k	.985	.997	.993	.968
U=3	20k	.925	1	.998	.949
U=4	10k	.956	1	1	.988
U=5	2k	1	1	1	.980

Table 3: Results for a model trained on five rules sharing the same predicate, and tested on multiple test sets.

6.4 Rule Chaining

Here, we assess models fine-tuned on various chaining depths. We construct six datasets for this scenario with increasing depths ($D = 0, D \leq 1, D \leq 2, D \leq 3, D \leq 4, D \leq 5$), i.e., dataset $D \leq x$ contains hypotheses that need at most x chained rules. We thus train six models (one per dataset), and we test them (i) on their own test dataset (Test), (ii) on the test set with $D \leq 5$ that contains all examples up to depth 5 (All), and (iii) on test sets with a chaining of depth x (Dep x).

Results. The results are shown in Table 4. We can see that the models achieve high F1 scores on the respective test sets for Depth 0. The red borderline indicates F1 scores for models tested on chaining depths higher than the ones they have been trained on. We see that Mod3 and Mod4 do fairly well on Depth 5. However, there is a decrease for higher depths, possibly due to the need for more training examples in order to learn such depths.

Data	Mod0	Mod1	Mod2	Mod3	Mod4	Mod5
Test	.996	.926	.883	.852	.856	.831
All	.589	.743	.772	.811	.831	.831
Dep0	.993	.974	.973	.982	.978	.973
Dep1	.264	.860	.884	.887	.889	.889
Dep2	.396	.655	.730	.751	.750	.720
Dep3	.438	.581	.636	.684	.690	.656
Dep4	.538	.468	.547	.626	.666	.627
Dep5	.552	.356	.496	.703	.785	.744

Table 4: F1 scores for models trained on varying depths and tested on six datasets. The boxed area indicates models tested on unseen chaining depths.

Moreover, since we sample a chain of rules each time, it is likely that every model has been trained on certain chains of rules. This yields lower scores in the constant-depth test sets as the models are being tested on unseen rule chains.

Note that Mod0 shows a counter-intuitive increase in the F1 score for higher unseen depths. Chaining soft rules may lead to a low probability for the associated hypothesis, and thus eventually to a *False* label. However, Mod0 is not trained on chaining and sees a hypothesis that requires chaining as an unsatisfied fact, thus eventually labelling it as *False*, while in fact it is the chaining of the soft rules that is the cause for this label. This is never the case with hard rules, as the actual label there would be *True*.

6.5 Testing RULEBERT on Unseen Rules

We have seen that a PLM can be successfully fine-tuned with rules. We now study the performance on the PLM after it has been fine-tuned on 161 (single) rules. We call this fine-tuned model RULEBERT.

	Rule	FT-PLM	RULEBERT ₂₀	FT-RULEBERT ₂₀
Known predicates	child(a,b) $\rightarrow$ parent(b,a)	.719	.869	.989
	relative(a,b) $\rightarrow$ negspouse(b,a)	.885	.885	.963
	child(a,b) $\wedge$ child(b,c) $\rightarrow$ negchild(a,c)	.835	.888	.918
	parent(a,b) $\wedge$ parent(a,c) $\rightarrow$ spouse(b,c)	.754	.757	.814
	parent(a,b) $\rightarrow$ negchild(a,b)	.923	.933	.963
Unknown predicates	knownFor(b,a) $\rightarrow$ founder(a,b)	.817	.795	.971
	worksFor(b,a) $\rightarrow$ negfounder(a,b)	.951	.915	.952
	occupation(a, b) $\rightarrow$ negalmaMater(a, b)	.939	.917	.972
	author(c,b) $\wedge$ series(a,c) $\rightarrow$ author(a,b)	.965	.937	.989
	city(a,b) $\rightarrow$ negstate(a,b)	.923	.912	.971

Table 5: Evaluation on unseen rules (accuracy). The first group contains rules with predicates seen by RULEBERT among the 20 rules used for fine-tuning, while the second group has rules with unseen predicates.

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child(a,b)  $\rightarrow$  negparent(a,b)
child(a,b)  $\rightarrow$  nespouse(a,b)
child(a,b)  $\rightarrow$  negchild(b,a)
child(a,b)  $\rightarrow$  negrelation(b,a)
parent(a,b)  $\rightarrow$  negparent(b,a)
parent(a,b)  $\rightarrow$  nespouse(a,b)
spouse(a,b)  $\rightarrow$  relative(b,a)
successor(a,b)  $\rightarrow$  predecessor(b,a)
predecessor(a,b)  $\rightarrow$  negsuccessor(a,b)
successor(a,b)  $\rightarrow$  negspouse(a,b)
predecessor(a,b)  $\rightarrow$  negspouse(a,b)
child(a,c)  $\wedge$  parent(c,b)  $\rightarrow$  spouse(a,b)
child(b,a)  $\wedge$  child(c,a)  $\rightarrow$  spouse(b,c)
parent(a,b)  $\wedge$  parent(b,c)  $\rightarrow$  negparent(a,c)
parent(a,b)  $\wedge$  child(c,a)  $\rightarrow$  spouse(b,c)
spouse(a,b)  $\wedge$  parent(c,a)  $\rightarrow$  negspouse(b,c)
spouse(a,b)  $\wedge$  child(a,c)  $\rightarrow$  negspouse(b,c)
successor(a,c)  $\wedge$  successor(b,c)  $\rightarrow$  negspouse(a,b)
publisher(c,b)  $\wedge$  subsequentwork(c,a)  $\rightarrow$  publisher(a,b)
publisher(c,b)  $\wedge$  previouswork(c,a)  $\rightarrow$  publisher(a,b)

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Figure 3: The 20 random rules used for RULEBERT₂₀.

We first evaluate RULEBERT on unseen rules. We fine-tune it with only twenty randomly selected rules (shown in Figure 3) and call it RULEBERT₂₀. We then select ten new rules divided into two groups: (i) five rules containing predicates that were used in the rules for fine-tuning RULEBERT₂₀, and (ii) five rules that share no predicates with the fine-tuning rules. For each rule in the test sets, we run a model fine-tuned (with 4k examples) only for that rule (FT-PLM), the model fine-tuned on the twenty original rules (RULEBERT₂₀), and the same model fine-tuned again for the rule at hand (FT-RULEBERT₂₀).

Results. Table 5 shows that RULEBERT₂₀ outperforms the fine-tuned model (FT-PLM) on the first group. Even though fine-tuned on 20 rules, it learned enough about (i) symmetric/transitive predicates and (ii) rule confidence to predict correctly, even better than rule-specific models.

For the second rule group, the accuracy of RULEBERT₂₀ is high, but FT-PLM performs better. Applying the same fine-tuning on RULEBERT₂₀ yields the best results in all scenarios.

7 RULEBERT on External Datasets

As our fine-tuning propagates information in the layers of the encoder, we hypothesize that RULEBERT effectively “learns” logical properties of the concepts represented in the rules, such as negation and symmetry, and thus it could perform better on tasks testing such properties of PLMs. To study the negation of predicates, we use the *Negated LAMA datasets*, which test how PLMs distinguish a Cloze question and its negation (Kassner and Schütze, 2020). In most cases, PLMs make the same prediction both for a positive statement (“*Relativity was developed by Einstein.*”) and for its negation (“*Relativity was not developed by Einstein.*”). To test the symmetry relationship between predicates, we use the SRL test in *CheckList* (Ribeiro et al., 2020), which focuses on behavioral testing of NLP models; we use its test set for the duplicate-question detection task (QQP) (Wang et al., 2019a). Finally, we test deductive reasoning on the *bAbI* dataset and its Task #15 (Weston et al., 2016).

7.1 Negated LAMA Experiments

For Negated LAMA, we do not fine-tune RULEBERT for the task; instead, we replace its original classification layer by an MLM head with weights identical to those of RoBERTa (not fine-tuned). Note that this configuration is biased in favor of RoBERTa, as the parameters of the MLM head and of the RoBERTa encoder have been trained in conjunction and thus good values have been found for this combination, which is not the case for our RULEBERT.

	Fine-Tuned	RoBERTa	RULEBERT
bAbI	1 epoch	.401	.477
	2 epochs	.676	.863
	3 epochs	.827	.825
Neg. LAMA	-	.684	.852
CheckList QQP	1 epoch	.000	.422
	3 epochs	.000	.000

Table 6: Evaluation on external datasets (accuracy).

Results Yet, even in this arguably unfair setting, RULEBERT outperforms RoBERTa on all datasets of Negated LAMA, as shown in Table 7. We can see that RULEBERT performs better on both evaluation measures used in (Kassner and Schütze, 2020). It achieves a lower mean Spearman rank correlation (ρ) and a much smaller percentage of positive and negated answers overlap (%).

7.2 CheckList QQP Experiments

The CheckList tests (Ribeiro et al., 2020) have shown that PLMs fail in many basic cases. We hypothesize that RULEBERT can perform better on tasks and examples that deal with symmetric and asymmetric predicates, if such predicates have been shown to it during pre-fine-tuning. We experiment with the QQP dataset, which asks to detect whether two questions are duplicates. We identify a few rules that can teach a model about symmetric predicates, and we pre-fine-tune RULEBERT on them; then, we fine-tune it on the QQP dataset.

Results Table 6 shows the results on the challenging CheckList QQP test set: we can see that RULEBERT achieves accuracy of 0.422 after one epoch, while RoBERTa is at 0.0. However, after three epochs RULEBERT is also at 0.0,³ i.e., it started to unlearn what it had learned at pre-fine-tuning (Kirkpatrick et al., 2017; Kemker et al., 2018; Biesialska et al., 2020). Learning a new task often leads to such catastrophic forgetting (Ke et al., 2021). While there are ways to alleviate this (Ke et al., 2021), this is beyond the scope of this paper.

7.3 bAbI Task #15 Experiments

Finally, we experiment with task #15 of the bAbI dataset, where the goal is to assess whether a model can perform deductive reasoning. However, as mentioned in the original bAbI paper (Weston et al., 2016), it is not only desirable to perform well on the task, but also to use the fewest examples.

³On the much easier QQP test set, RULEBERT achieved 0.89 accuracy after one epoch, and 0.91 after three epochs.

		Facts	RoBERTa		RULEBERT	
			ρ	%	ρ	%
GR	birthplace	2,404	90.99	18.51	71.72	4.20
	birthdate	1,565	82.87	1.40	63.55	0.13
	deathplace	649	86.44	0.31	71.13	0.00
T-REx	1-1	973	78.95	61.38	51.21	32.96
	N-1	20,006	87.56	43.80	67.63	11.48
	N-M	13,096	89.39	50.78	72.59	28.90
ConceptNet	—	2,996	42.61	9.00	37.43	4.83
SQ	—	286	89.71	44.76	75.05	26.32

Table 7: Negated LAMA: Mean Spearman rank correlation (ρ) and mean percentage of overlap in the first ranked predictions (%) for original vs. negated queries.

Thus, we use the smallest dataset consisting of about 2,000 data points. We hypothesize that under the same conditions and hyper-parameters, RULEBERT should be able to generalize faster and to learn in fewer epochs. As PLMs produce varying scores when fine-tuned on small datasets, we repeat the experiment ten times and we report the average scores. We then compare to RoBERTa. Both models contain two classification layers to predict start and end spans of the input context.

Results We can see in Table 6 that RULEBERT achieves accuracy of 0.863 in two epochs, while RoBERTa achieves 0.676. On the third epoch, RoBERTa catches up with accuracy of 0.827, while RULEBERT starts to overfit (goes down to 0.825), indicating that fewer epochs should be used, probably due to catastrophic forgetting.

8 Conclusion and Future Work

We studied whether PLMs could reason with soft rules over natural language. We experimented with one flavor of probabilistic answer set programming (LP^{MLN}), but other semantics can be also used with the proposed methodology. We further explored the inference capabilities of Transformer-based PLMs, focusing on positive and negative textual entailment.

We leave non-entailment for future work. We also leave open the development of explainable models. Some approaches use occlusion that removes parts of the input and checks the impact on the output (Clark et al., 2020) or build proof iteratively using 1-hop inference (Tafjord et al., 2021).

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